

COLUMBUS AFB, MS, USA

WMO: 723306

Lat: **33.650N**Lon: **88.450W**Elev: **219**StdP: **14.58**Time Zone: **-6.00 (NAC)**Period: **94-19**WBAN: **13825****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	20.7	24.6	7.4	8.1	27.8	12.1	10.3	30.3	24.0	49.7	21.4	45.3	6.7	340	0.410

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	19.6	96.4	76.2	94.2	76.7	92.4	76.4	81.3	89.9	80.0	89.1	78.7	87.7	7.3	280

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
79.2	152.5	85.7	77.5	143.9	83.7	76.6	139.6	82.9	44.9	89.8	43.5	89.0	42.3	88.0	87.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 19.5	17.3	15.4	15.0	98.8	5.2	3.0	11.3	101.0	8.2	102.7	5.3	104.4	1.5	106.5		
(5)			13.5	82.5	5.1	1.9	9.9	83.9	6.9	85.0	4.1	86.1	0.4	87.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	63.8	43.7	49.3	55.7	63.4	71.8	78.7	81.2	80.9	75.5	64.4	52.5	47.5
	DBStd	15.31	10.87	10.60	9.52	7.26	6.08	4.08	3.47	3.69	5.84	8.07	8.58	9.79
	HDD50	691	260	131	54	3	0	0	0	0	0	3	71	169
	HDD65	2599	664	446	308	116	18	0	0	0	4	114	381	548
	CDD50	5725	63	112	232	405	676	861	968	956	766	448	147	91
	CDD65	2158	2	8	21	67	229	411	503	491	320	94	7	5
	CDH74	21546	2	47	211	670	2141	4135	5260	5002	3073	936	57	12
	CDH80	9568	0	2	31	142	788	1929	2543	2459	1380	288	6	0
(14) Wind	WSAvg	6.3	7.8	7.9	7.9	7.5	6.1	5.3	4.8	4.5	4.9	5.6	6.0	7.0
(15) Precipitation	PrecAvg	56.0	5.5	5.2	5.8	5.6	4.4	4.3	4.7	3.7	3.4	3.1	4.2	5.9
	PrecMax	74.4	10.6	8.8	12.1	14.3	13.6	8.9	10.3	7.6	10.5	10.9	8.3	12.5
	PrecMin	32.0	2.6	1.7	0.7	2.4	1.1	0.4	1.1	0.8	0.1	0.0	1.8	1.7
	PrecStd	10.0	2.0	2.0	2.1	2.8	2.9	2.4	2.4	1.9	2.2	2.4	1.6	2.5
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	72.9	79.0	83.4	86.8	91.9	96.9	98.4	100.6	96.8	90.3	79.7	75.7
		MCWB	63.4	65.9	65.8	69.0	73.3	76.0	77.0	74.8	72.8	71.3	66.2	68.1
	2%	DB	69.2	74.9	79.6	83.9	89.8	94.2	95.4	96.7	93.4	86.3	74.8	71.2
		MCWB	62.3	64.3	63.9	67.7	72.7	76.1	78.3	75.7	74.2	70.6	63.9	65.5
	5%	DB	65.5	70.8	75.8	81.2	87.5	91.9	93.4	93.6	91.0	83.2	71.4	67.6
		MCWB	60.2	61.4	62.4	66.4	72.1	75.9	77.9	76.9	74.5	68.5	63.3	63.1
	10%	DB	61.7	66.4	72.1	78.1	85.0	89.9	91.3	91.2	88.3	79.6	67.9	63.7
		MCWB	56.1	58.9	60.5	64.5	71.0	75.4	77.5	76.9	73.6	66.7	60.6	59.7
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	67.0	69.9	70.1	73.3	77.2	80.9	83.3	83.5	79.1	76.9	71.4	69.5
		MCDB	70.1	74.8	77.0	81.3	86.2	89.8	92.1	91.5	88.0	82.9	74.2	73.7
	2%	WB	64.3	66.7	67.4	71.1	75.4	79.4	81.3	81.6	77.4	74.3	68.2	66.5
		MCDB	66.9	72.1	76.3	79.1	84.8	88.8	90.2	90.0	87.2	82.2	71.5	69.7
	5%	WB	61.5	64.0	65.2	69.2	74.1	78.3	80.1	80.0	76.3	72.2	64.8	63.8
		MCDB	64.5	68.8	72.1	76.9	83.6	87.6	89.5	89.0	85.8	79.1	68.9	66.1
	10%	WB	57.9	60.8	63.1	67.5	72.9	77.2	78.9	78.4	75.1	69.8	62.0	60.3
		MCDB	61.1	65.0	69.1	75.2	81.8	86.2	88.1	86.8	84.1	75.6	66.5	63.3
(35) Mean Daily Temperature Range	5% DB	MDBR	19.8	21.2	22.8	23.6	22.0	20.4	19.6	20.1	22.3	24.4	23.0	19.3
		MCDBR	22.2	24.8	26.8	25.7	24.5	23.1	22.2	23.6	25.3	26.9	26.2	21.2
		MCWBR	16.1	14.8	14.0	12.1	9.8	8.3	7.9	7.7	8.9	12.0	16.0	15.1
		MCDBR	17.6	19.9	21.6	20.4	20.6	19.8	19.8	19.2	20.4	19.9	20.3	18.3
	5% WB	MCWBR	15.0	13.8	13.9	11.6	9.2	8.5	8.7	8.5	8.3	10.7	15.4	14.8
(40) Clear-Sky Solar Irradiance	taub		0.322	0.328	0.364	0.397	0.433	0.479	0.515	0.515	0.436	0.369	0.341	0.319
	taud		2.513	2.491	2.413	2.317	2.260	2.151	2.082	2.076	2.286	2.453	2.472	2.537
	Ebn at Noon		283	293	289	283	272	259	249	246	262	275	273	278
	Edh at Noon		27	30	35	41	43	49	52	51	40	31	28	25
(44) All-Sky Solar Radiation	RadAvg		826	984	1351	1718	1906	2016	1938	1777	1567	1275	934	705
	RadStd		69	136	127	124	134	146	108	119	133	165	94	69

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (26 neighbors)	+0.79	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+243	+171